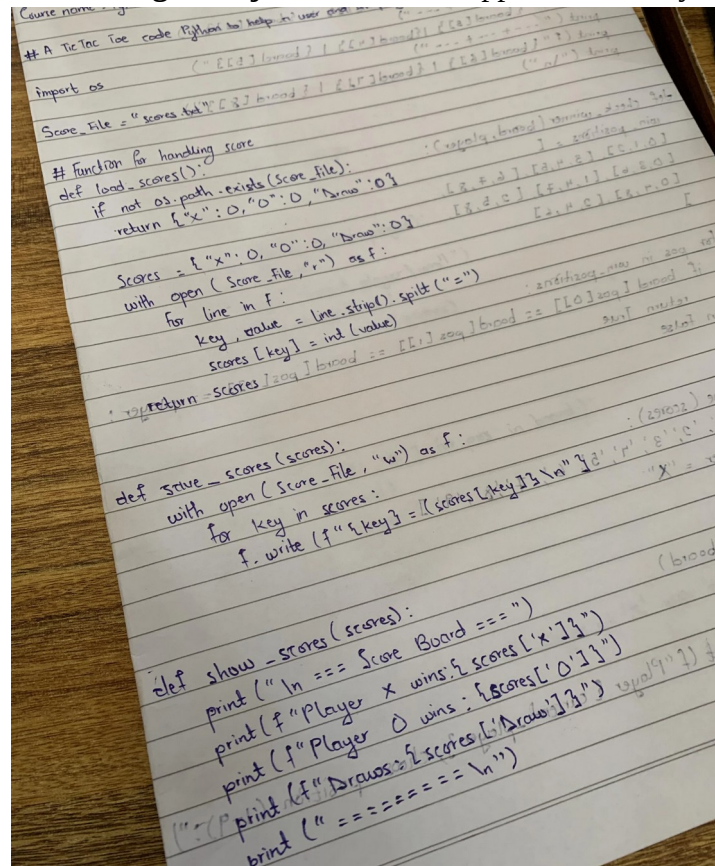


Software Requirement Specification

By **Coding** กระดาษ

Chosen Irl Challenge: Project D: Course Support & Activity Dashboard



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1. Target Users

- **TAs:** View and manage their own recurring tasks (lab prep, consultation slots, grading deadlines), log issues/FAQs encountered during support, and track progress on assigned milestones. For New TAs: View a structured overview of the course's recurring rhythm (what typically happens each week) to quickly understand expectations without needing to ask senior staff.
- **Lecturers:** Oversee the full course timeline across all TAs, see which tasks/deadlines are at risk, and monitor how student support activities (consultations, lab help, FAQs) are trending over the semester.
- **Students:** Post or view FAQ/issue entries related to course activities, without needing full account access to task or schedule data.

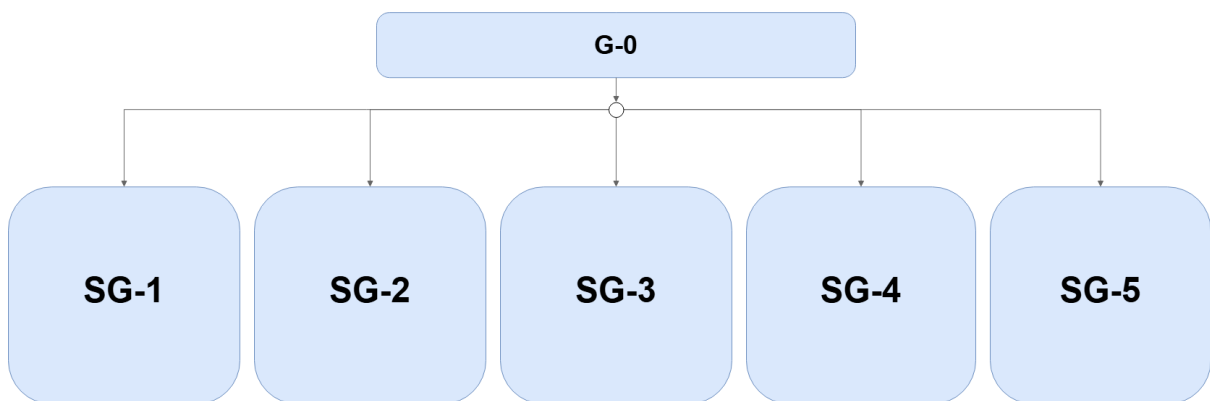
2. The System Goal

G-0: Maintain an integrated, up-to-date view of course activities and support workload.

Aj. Suwichak's department and Aj. Kundjanasith's department currently relies on fragmented tools — spreadsheets, calendar entries, chat groups, and email threads — to coordinate assignment deadlines, lab sessions, consultation slots, project milestones, and internal tasks. This makes it hard to see "what's happening when," which tasks are at risk, and how student support activity is evolving over the semester. New TAs in particular struggle to learn the course rhythm from scattered sources.

For this iteration, the system scope is limited to **activity tracking** (tasks, deadlines, milestones — administrator/TA-facing) and **support logging** (consultations, lab help, FAQs — TA-facing, viewable by lecturers). Multi-platform chat/email scraping, automated notifications, and grading/submission integrations are explicitly out of scope for this iteration and will be addressed in later iterations.

3. Goal Refinement via KAOS:



G-0: Maintain an integrated, up-to-date view of course activities and support workload.

- **Sub-Goal 1 (SG-1)**

Description: Achieve: Provide users with a centralized view of course activities.

Rationale: TAs and lecturers currently rely on scattered spreadsheets, calendars, chats, and emails to find activity information.

- **Sub-Goal 2 (SG-2)**

Description: Maintain: Complete and up-to-date activity information.

Rationale: Each activity needs accurate information such as title, date, time, type, responsible person, and status so that users know what needs to be done and when.

- **Sub-Goal 3 (SG-3)**

Description: Achieve: Allow authorised users to create and update activities.

Rationale: Lecturers and authorised TAs need to maintain course information while preventing unauthorised changes.

- **Sub-Goal 4 (SG-4)**

Description: Avoid: Missed, conflicting, or outdated activities.

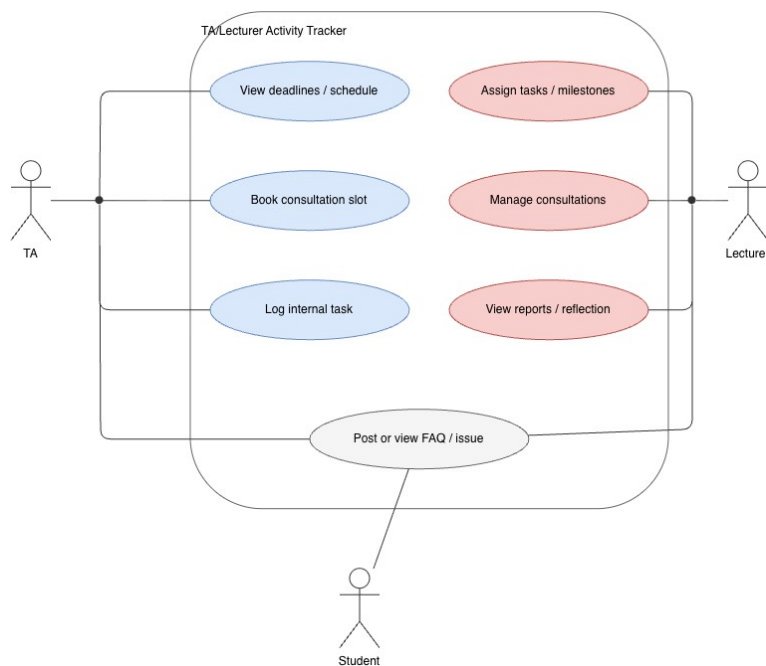
Rationale: Fragmented information can cause deadlines, lab sessions, consultation slots, and other responsibilities to be overlooked or incorrectly communicated.

- **Sub-Goal 5 (SG-5)**

Description: Maintain: A manageable history of course activities.

Rationale: Completed and archived activities should remain available for reporting, reflection, future planning, and reference.

4. Use Case Diagram



5. Use Stories

User Story ID	Description	Acceptance Criteria
US-1	As a TA, I want to see all upcoming deadlines, lab sessions, consultations and milestones in one view so that I don't have to check several tools.	The dashboard filters and displays active activity details (title, type, schedule, location, and assignee) based on the selected course while automatically hiding past items.
US-2	As a lecturer, I want to create a course activity so that all TAs work from the same schedule.	Creating an activity with all required details immediately saves and syncs it across the shared schedule for all assigned TAs.
US-3	As a lecturer, I want to assign a task to a specific TA so that responsibility is clear.	Assigning a task to an active TA instantly displays it in their personal task list while logging the assignment timestamp.
US-4	As a TA, I want to update the status of my task so that the team knows what has been completed.	Only the assigned TA or the lecturer can change status. The change is timestamped and reflected on the dashboard after saving.
US-5	As a lecturer, I want overdue and at-	A task past its due date and not marked

	risk tasks flagged automatically so that I can act before a deadline is missed.	Done is flagged Overdue; a task due within the configured warning period is flagged At Risk. Flags appear on the dashboard.
US-6	As a TA, I want to record a consultation session so that consultation load is documented rather than forgotten.	A session record stores date, duration, topic and student/group. Saved records appear in the TA's history.
US-7	As a TA, I want to search and submit FAQ/issue entries so that recurring student questions are answered consistently.	Keyword search covers question text, answer and tags. A submitted entry is saved as Pending and becomes visible to others only after lecturer approval.
US-8	As a lecturer, I want access controlled by role so that only authorised users can change course data.	Users must authenticate before accessing course data. TAs cannot create courses or approve FAQ entries; unauthorised attempts are rejected.
US-9	As a lecturer, I want to view completed and archived activities from a past period so that I can review how support load changed over the semester.	Selecting a course and date range retrieves the completed and archived records for that period and displays them as an activity and support-load summary, while showing a 'no records' message when none exist.
US-10	As a student, I want to search and view published FAQ/issue entries, and submit a new question when I can't find an answer, so that I get a consistent answer without waiting for direct contact.	Keyword search covers question text, answer and tags for entries with status Published. A student-submitted question is saved as Pending and becomes visible only after lecturer approval; Pending and Rejected entries remain hidden from students.
US-11	As a TA, I want to sign in with my existing university account so that I do not have to manage a separate password for this system.	Selecting "Sign in with university account" redirects the user to the identity provider. On successful authorisation the system creates a session, resolves the user's role, and returns to the dashboard. Accounts that are not registered course members are denied access.

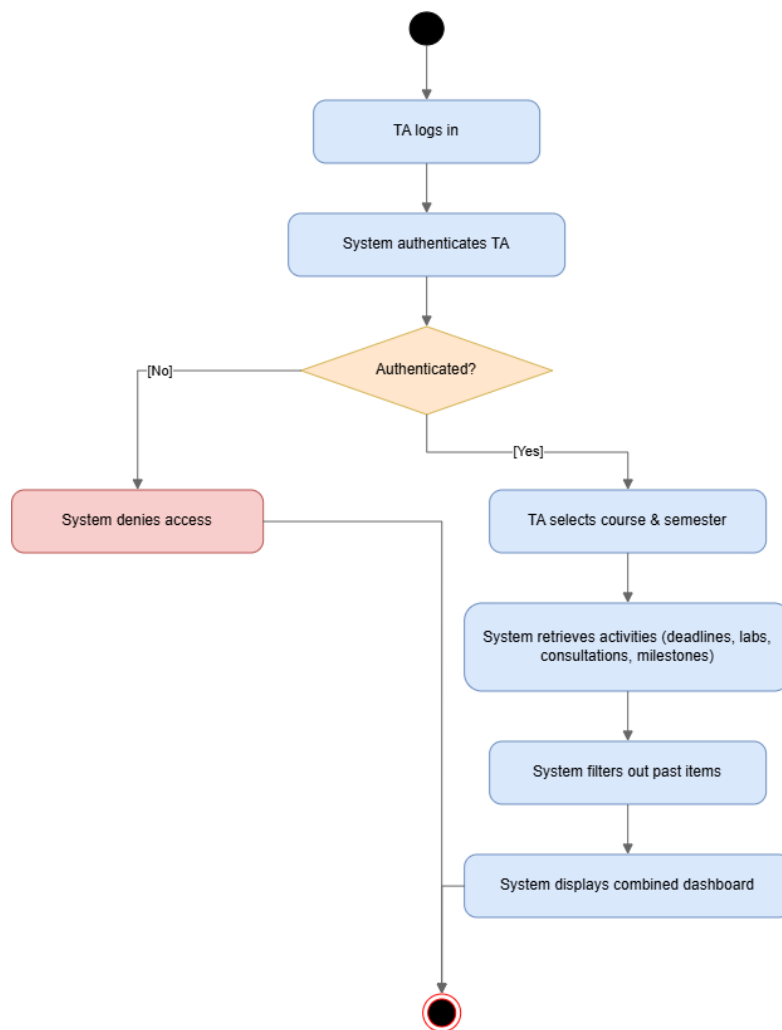
6. User Requirement Specification

ID	User Requirement
URS-1	Users shall be required to sign in through an external OAuth 2.0 / OpenID Connect identity provider (institutional single sign-on) before accessing any course data, and shall be granted access according to their assigned role.
URS-2	The system shall provide a single combined view of deadlines, lab sessions, consultation slots and milestones for a selected course and semester.
URS-3	Users shall be able to view each activity's essential information, including title, type, date and time, location, assignee and status.
URS-4	The system shall allow an authorised lecturer to create, edit and remove course activities, and to assign a task to a specific TA.
URS-5	The system shall allow an assigned TA to update the status of their own tasks.
URS-6	The system shall automatically indicate tasks that are overdue or approaching their due date.
URS-7	The system shall allow a TA to record a consultation session, including date, duration, topic and participants.
URS-8	The system shall provide a searchable FAQ/issue log in which new entries require lecturer approval before publication.
URS-9	The system shall restrict activity management and FAQ/issue approval actions to users holding the appropriate role, and shall reject such actions from unauthorised users.
URS-10	The system shall allow a lecturer to view completed and archived course activities and support-log records for a selected course and date range for reporting and reflection.

7. Activity Diagram

AD-1: TA Views Combined Dashboard

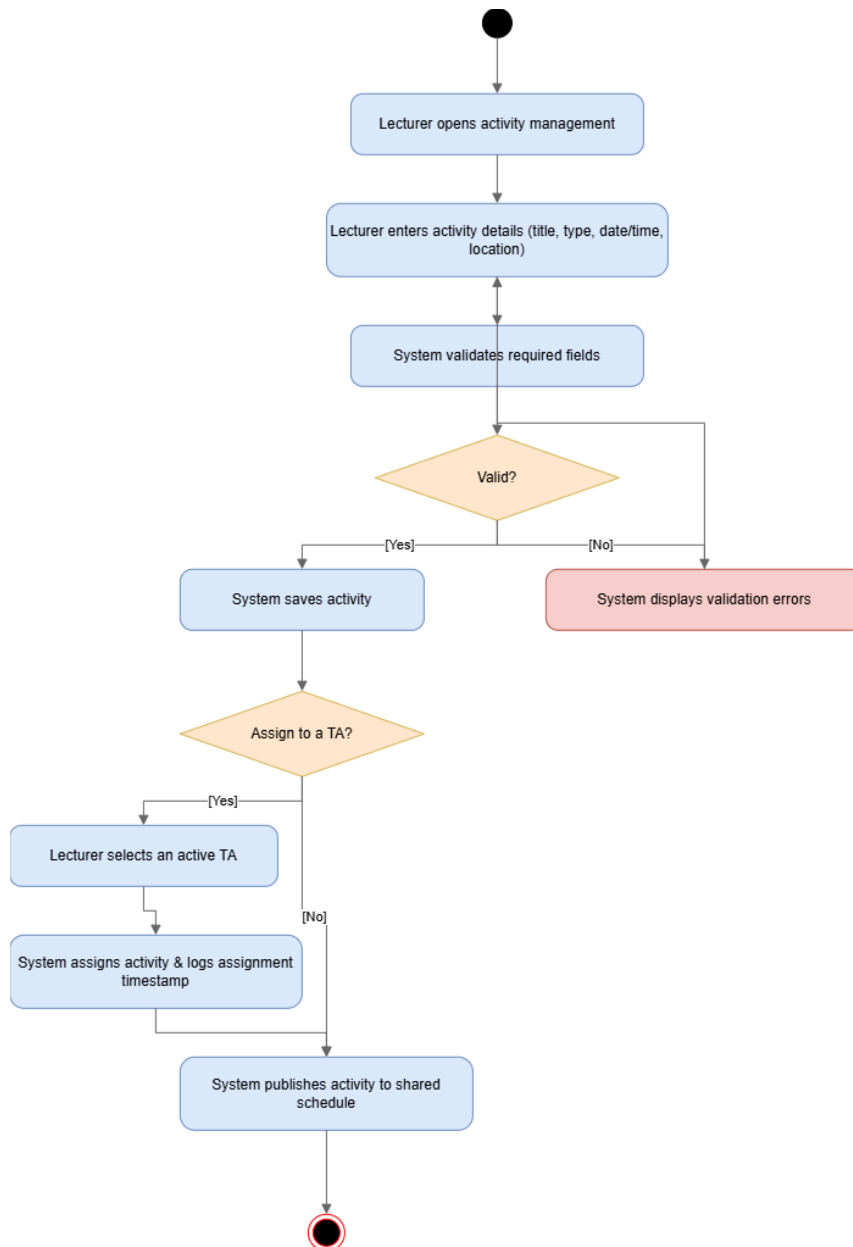
This diagram traces the login and dashboard-retrieval flow used to derive SRS-1, SRS-2, SRS-3 and SRS-4. It realises SG-1 (Centralized Activity View).



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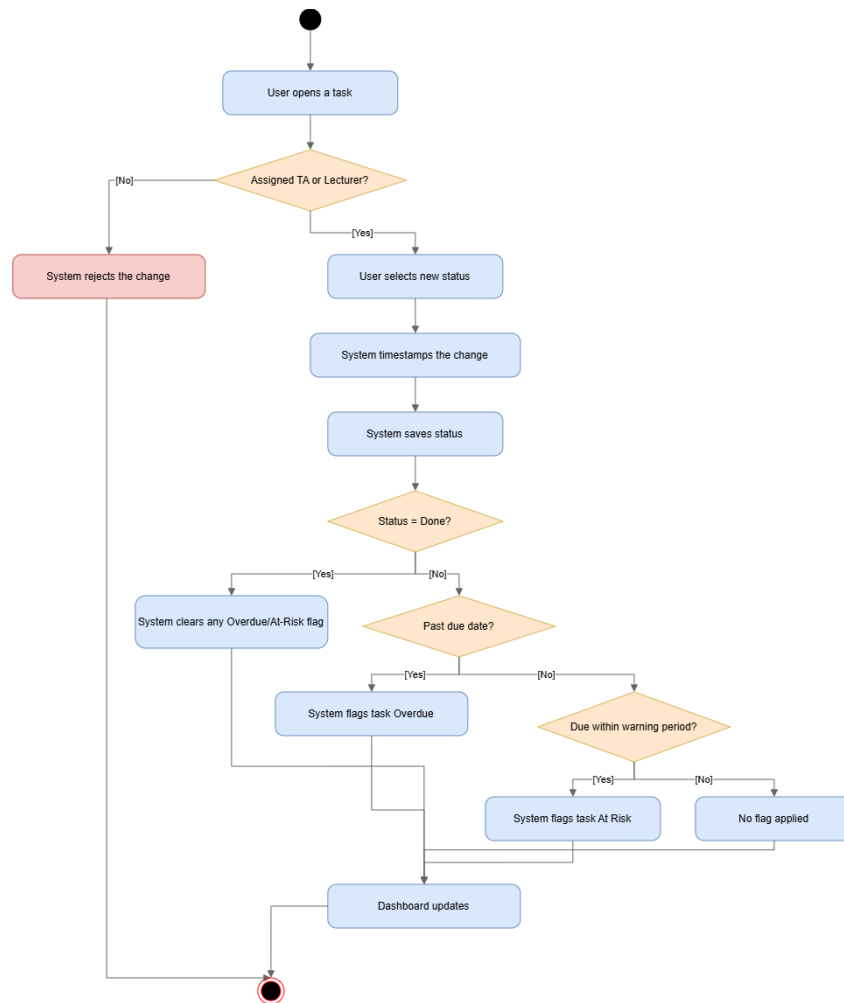
AD-2: Lecturer Creates & Assigns an Activity

This diagram was used during elicitation to discover validation rules and the assignment flow, which were then converted into SRS-5 through SRS-9. It realises SG-2 (Accurate Activity Information) and SG-3 (Controlled Activity Management).



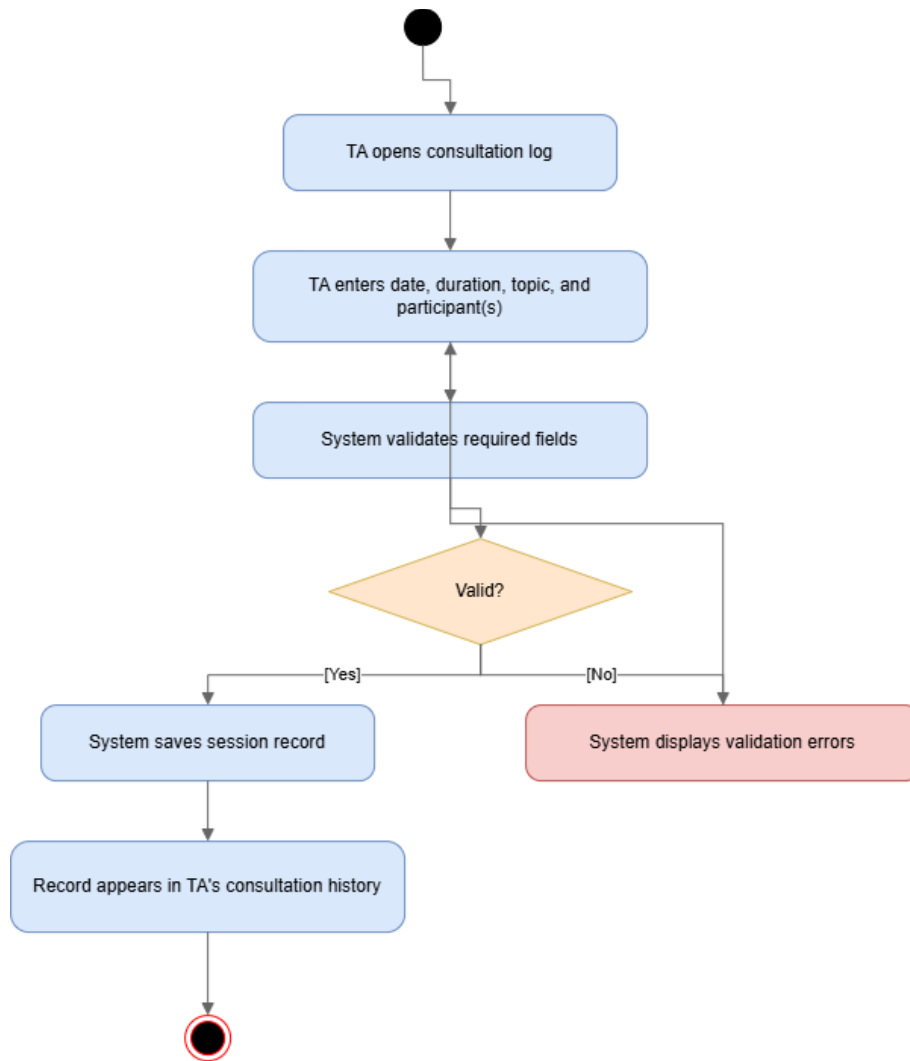
AD-3: TA Updates Task Status (with Overdue / At-Risk Check)

This diagram captures the authorisation check on status changes and the automatic overdue/at-risk flagging logic used to derive SRS-10 through SRS-13. It realises SG-3 (Controlled Activity Management) and SG-4 (Activity Conflict Prevention).



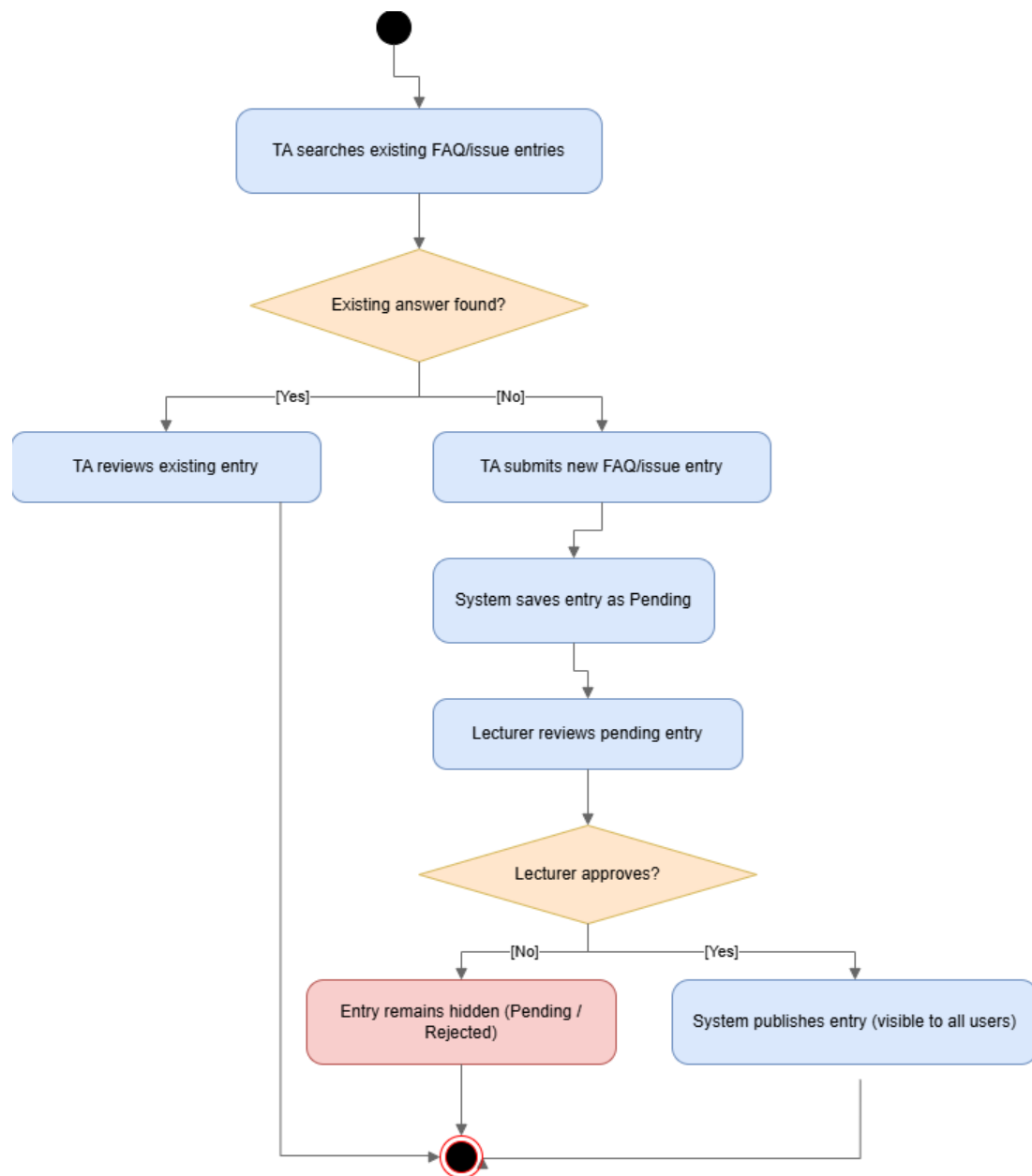
AD-4: TA Records a Consultation Session

This diagram traces the consultation-logging flow used to derive SRS-14 and SRS-15. It realises SG-2 (Accurate Activity Information).



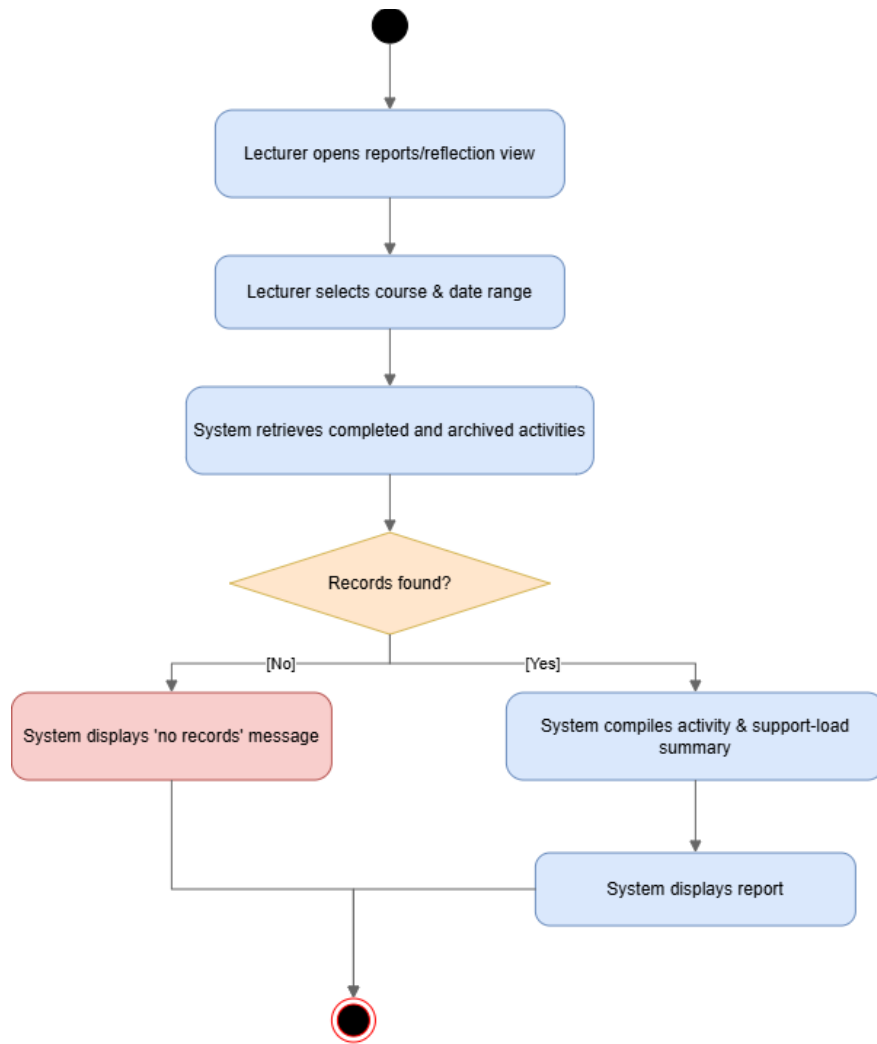
AD-5: TA Submits FAQ / Issue for Lecturer Approval

This diagram captures the search-first, pending-approval workflow used to derive SRS-16 through SRS-18. It realises SG-3 (Controlled Activity Management) and SG-4 (Activity Conflict Prevention), since unreviewed answers are kept hidden until an authorised lecturer approves them.



AD-6: Lecturer Views Reports & Activity History

This diagram captures the retrieval of completed and archived records used to derive SRS-19 and SRS-20. It realises SG-5 (Activity History Management).



8. System Requirement Specification

ID	System Requirement
SRS-1	The system shall require users to authenticate via an external OAuth 2.0 / OpenID Connect identity provider before displaying any course data.
SRS-2	The system shall provide a single combined view of deadlines, lab sessions, consultation slots and milestones for the selected course and semester.
SRS-3	The system shall display each activity's title, type, date and time, location, assignee, and status.
SRS-4	The system shall automatically exclude past (completed) activities from the default dashboard view.
SRS-5	The system shall allow only an authorised lecturer to create, edit, or remove a course activity.
SRS-6	The system shall require a title, type, date/time, and location before saving a new or edited activity.
SRS-7	The system shall display a validation message and shall not save the activity when required fields are missing or invalid.
SRS-8	The system shall allow a lecturer to assign an activity or task to a specific active TA and log the assignment timestamp.
SRS-9	The system shall immediately sync a saved activity to the shared schedule of all assigned TAs.
SRS-10	The system shall allow only the assigned TA or the lecturer to change a task's status; changes attempted by any other user shall be rejected.
SRS-11	The system shall timestamp every status change and reflect it on the dashboard after saving.
SRS-12	The system shall flag a task as Overdue when its due date has passed and its status is not Done.
SRS-13	The system shall flag a task as At Risk when it is due within the configured warning period and its status is not Done.
SRS-14	The system shall allow a TA to record a consultation session with date, duration, topic, and participant(s), and require all fields before saving.
SRS-15	The system shall display a saved consultation session in the recording TA's history.
SRS-16	The system shall provide keyword search over FAQ/issue question text, answer, and tags.
SRS-17	The system shall save a newly submitted FAQ/issue entry with a status of Pending and make it visible to other users only after an authorised lecturer approves it.
SRS-18	The system shall reject an FAQ/issue approval attempt, or a course-activity management attempt, made by a user without the required authorisation.
SRS-19	The system shall allow a lecturer to view completed and archived activities for a selected course and date range.
SRS-20	The system shall retain completed and archived activity and support-log records for reporting and future reference rather than deleting them.

SRS-21	The system shall use the OAuth 2.0 Authorization Code flow to obtain an access token and an ID token from the identity provider, and shall not store or handle user passwords.
SRS-22	The system shall map the authenticated account (provider subject identifier and institutional email) to an internal user record holding the role TA, Lecturer or Student, and shall use that role for all subsequent authorisation checks.
SRS-23	The system shall deny access and display an authentication error when the returned token is missing, expired or invalid, or when the authenticated account is not registered as a member of the selected course.

9. Software Architecture

Sub-Goal	Use Case	User Story	URS	Activity Diagram	SRS
SG-1	View deadlines / schedule	US-1	URS-1, URS-2, URS-3	AD-1	SRS-1, SRS-2, SRS-3, SRS-4
SG-2	Assign tasks / milestones; Manage consultations	US-2, US-6	URS-4, URS-7	AD-2, AD-4	SRS-6, SRS-7, SRS-9, SRS-14, SRS-15
SG-3	Assign tasks / milestones; Log internal task; Post or view FAQ / issue	US-2, US-3, US-4, US-8, US-11	URS-1, URS-4, URS-5, URS-8	AD-2, AD-3, AD-5	SRS-1, SRS-5, SRS-8, SRS-10, SRS-11, SRS-16, SRS-17, SRS-18, SRS-21, SRS-22, SRS-23
SG-4	Log internal task; Post or view FAQ / issue	US-5, US-7	URS-6, URS-8	AD-3, AD-5	SRS-12, SRS-13, SRS-17, SRS-18
SG-5	View reports / reflection	—	—	AD-6	SRS-19, SRS-20

10. Data Storage Technology

The recommended data storage technology is a Relational Database Management System (RDBMS), such as PostgreSQL or MySQL, used as the single data store for this iteration.

Rationale

All data described in the SRS is structured and fits a fixed, well-defined schema: every activity has a title, type, date/time, location, assignee and status (SRS-3); every task carries a status, timestamp and Overdue/At-Risk flag (SRS-10 to SRS-13); every consultation session has a date, duration, topic and participants (SRS-14); and every FAQ/issue entry has question text, an answer, tags and an approval status (SRS-16 to SRS-18). These entities are related to one another (a Lecturer creates many Activities, a TA is assigned many Tasks and logs many Consultations, a User submits many FAQ entries), which favours a relational model with foreign keys over a document or key-value store.

An RDBMS also enforces the integrity and authorisation constraints the SRS requires: rejecting a status change from an unauthorised user (SRS-10, SRS-18), keeping submitted FAQ entries hidden until approved (SRS-17), and reliably filtering active data from completed or archived records for reporting (SRS-4, SRS-19, SRS-20). Transactional consistency ensures that a status update, its timestamp, and any resulting Overdue/At-Risk flag are saved together rather than partially. Given the modest data volume and a small team, a relational database keeps the technology stack simple to operate and query, avoiding the added operational complexity of a separate NoSQL engine for no functional benefit at this stage.

Component	Data Storage Technology	Description
Activities table	RDBMS (e.g., PostgreSQL/MySQL)	Stores deadlines, lab sessions, consultation slots and milestones with title, type, date/time, location, assignee and status (SRS-2, SRS-3, SRS-4).
Tasks table	RDBMS	Stores task status, assignment timestamp, and computed Overdue/At-Risk flags (SRS-8, SRS-10–SRS-13).
Consultation Sessions table	RDBMS	Stores date, duration, topic and participant(s) per TA (SRS-14, SRS-15).
FAQ/Issue table	RDBMS	Stores question, answer, tags and status (Pending / Published / Rejected) (SRS-16–SRS-18).
Users & Roles table	RDBMS	Stores the OAuth provider name, provider subject identifier, institutional email, display name and role (TA / Lecturer / Student) used for authorisation checks. No passwords are stored, since credential verification is delegated to the identity provider (SRS-1, SRS-18, SRS-21–SRS-23).

Only one data store is required for the current scope, since activity tracking and support logging both operate on structured, relationally-linked entities. Should a later iteration introduce unstructured data (for example, free-form chat/email scraping, which is explicitly

out of scope for this iteration), a supplementary document store could be added at that time without disrupting the relational core.

11. Development Environment

The recommended development and deployment environment is a containerised environment using Docker and Docker Compose, rather than bare-metal or a single shared VM without containers.

Rationale

The system must be usable by TAs, lecturers, new TAs and students who each access it from their own machines, and the project is developed by a multi-person team working in parallel across several modules (activity tracking, task assignment, consultation logging, FAQ approval). Docker packages the application and its dependencies into a portable, reproducible image, so every team member runs an identical environment regardless of their own laptop's operating system or installed software — removing the classic "it works on my machine" failure mode during integration.

Docker Compose additionally lets the application container and the RDBMS container be started together with one command, which keeps local setup fast for new contributors and mirrors the eventual deployment environment closely, reducing surprises when the system is deployed to a shared or hosted server for the course. Running the database in its own container also isolates it from the host machine and makes it easy to reset test data during development without affecting anyone else's environment.

Because authentication is delegated to an external OAuth 2.0 / OpenID Connect identity provider, the OAuth client ID, client secret and redirect URI are supplied to the application container as environment variables through Docker Compose rather than being committed to the repository. Each team member therefore develops against a separate development OAuth client with a local redirect URI, while the deployed instance uses its own client credentials and hosted redirect URI, so no real credentials are shared through version control.

Component	Development Environment	Description
Application Server (Controller + Model)	Docker container	Runs the backend application logic in an isolated, reproducible image that behaves identically on any team member's machine or the deployment server.
Database (RDBMS)	Docker container (via Docker Compose)	Runs the relational database in its own container, orchestrated with the application container, keeping data storage decoupled from host configuration.
Local Development Machine	Docker Desktop / local VM	Used by each team member to build and test their module before merging, mirroring the shared deployment configuration.

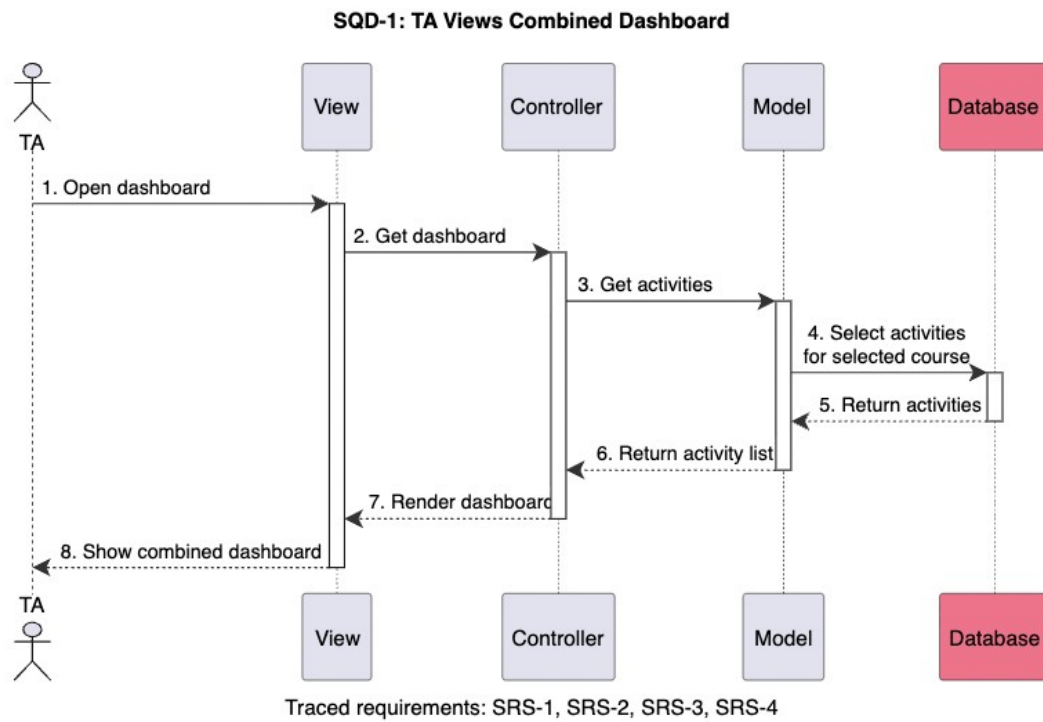
12. Sequence Diagrams for All SRS

The following sequence diagrams are derived from a Modular Monolithic (MVC) architecture — View, Controller, Model and Database layers — and cover the core flows traced in the SRS: viewing the combined dashboard, creating and assigning an activity, updating task status, recording a consultation, submitting and approving an FAQ/issue entry, and viewing reports. Each diagram lists the SRS items it traces, matching the numbering used in the User Story / URS / Activity Diagram traceability matrix.

SQD-1: TA Views Combined Dashboard

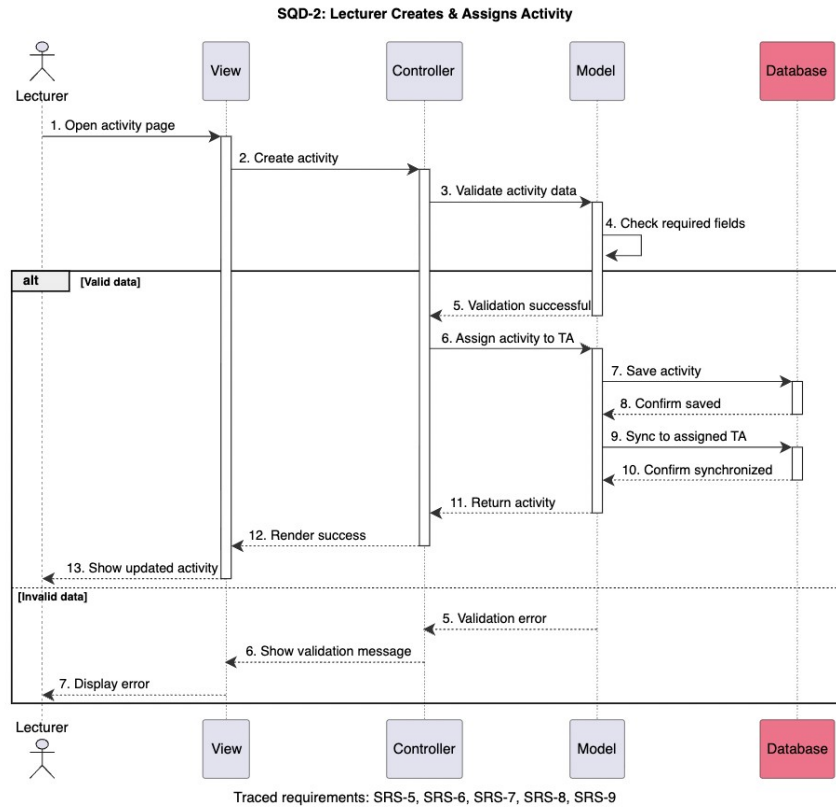
Traced requirements: SRS-1, SRS-2, SRS-3, SRS-4

Step 1 expands into the OAuth 2.0 Authorization Code flow: the View redirects the TA to the identity provider, the provider returns an authorisation code to the Controller, the Controller exchanges that code for an access/ID token, and the Model resolves the provider subject identifier to an internal user record and role before the dashboard is retrieved (SRS-1, SRS-21, SRS-22, SRS-23).



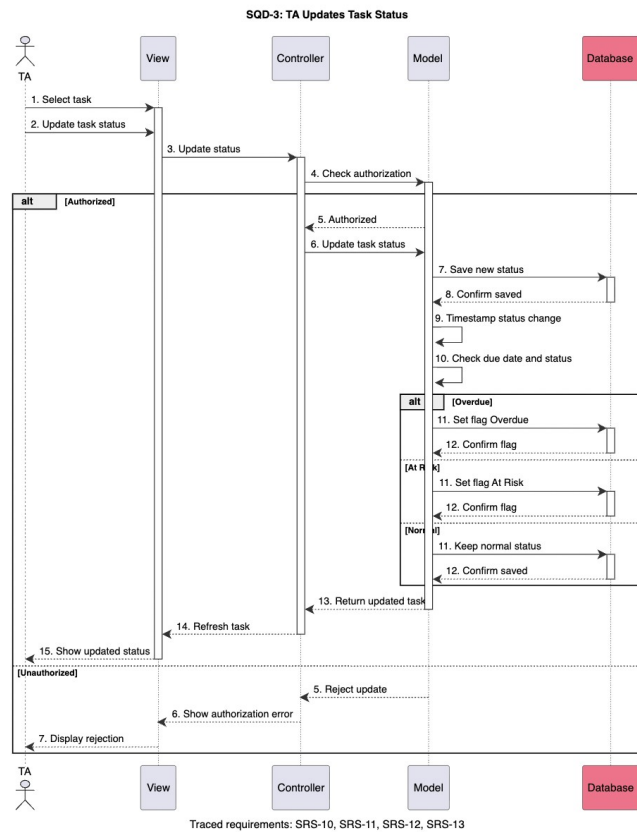
SQD-2: Lecturer Creates & Assigns Activity

Traced requirements: SRS-5, SRS-6, SRS-7, SRS-8, SRS-9



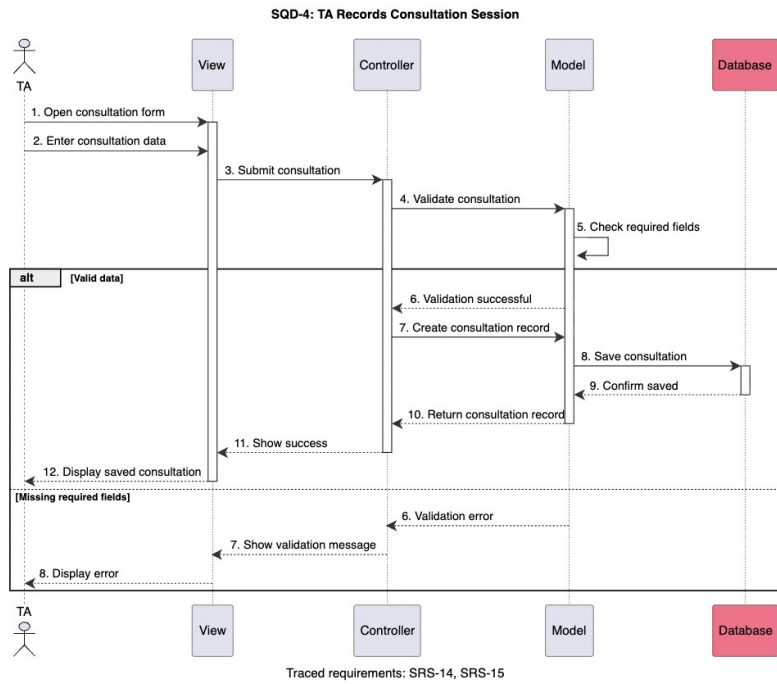
SQD-3: TA Updates Task Status

Traced requirements: SRS-10, SRS-11, SRS-12, SRS-13



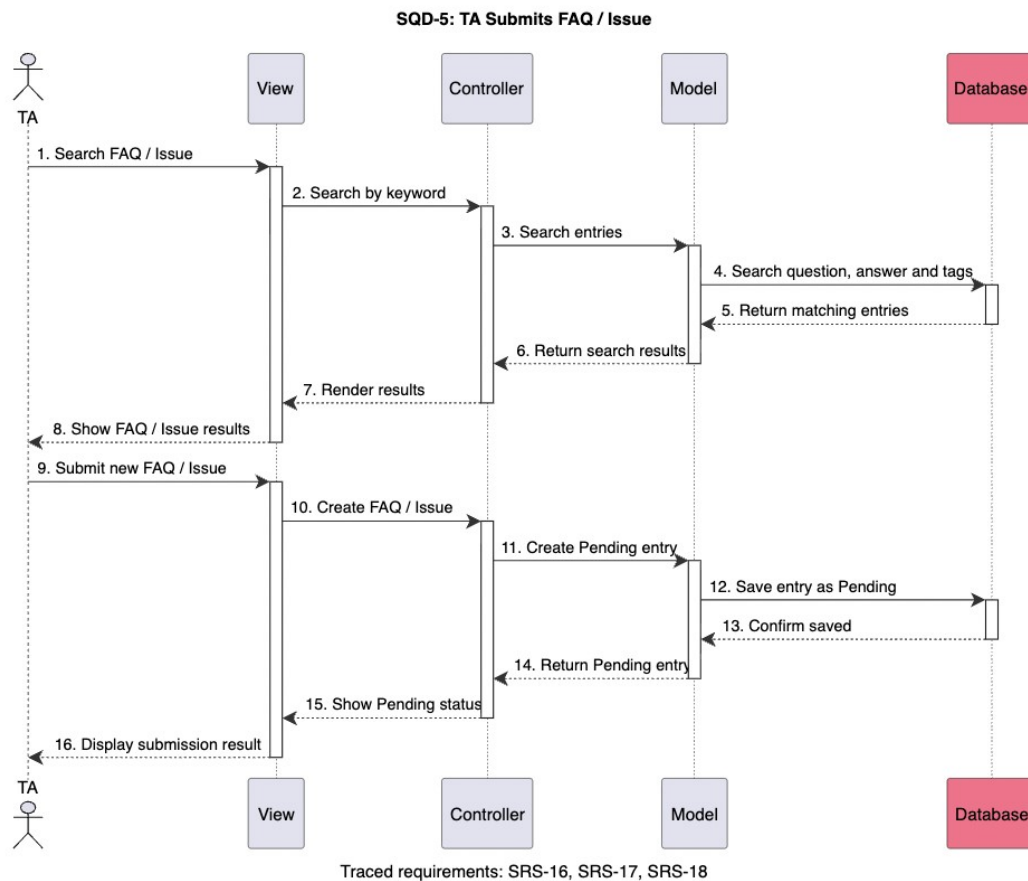
SQD-4: TA Records Consultation Session

Traced requirements: SRS-14, SRS-15



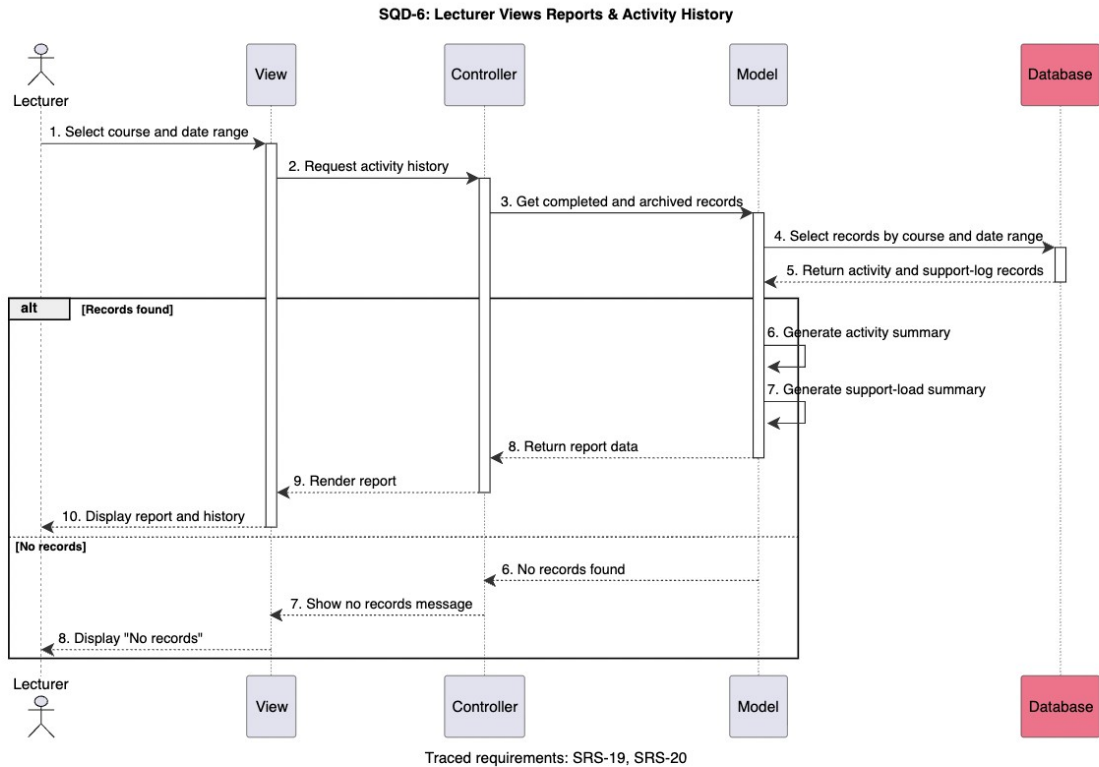
SQD-5: TA Submits FAQ / Issue

Traced requirements: SRS-16, SRS-17, SRS-18



SQD-6: Lecturer Views Reports & Activity History

Traced requirements: SRS-19, SRS-20



13. Traceability Matrix

From Sub-Goal to SRS

Sub-Goal	Use Case	User Story	URS	Activity Diagram	SRS
SG-1	View deadlines / schedule	US-1	URS-1, URS-2, URS-3	AD-1	SRS-1, SRS-2, SRS-3, SRS-4
SG-2	Assign tasks / milestones; Manage consultations	US-2, US-6	URS-4, URS-7	AD-2, AD-4	SRS-6, SRS-7, SRS-8, SRS-9, SRS-14, SRS-15
SG-3	Assign tasks / milestones; Log internal task; Post or view FAQ / issue	US-2, US-3, US-4, US-8, US-11	URS-1, URS-4, URS-5, URS-8	AD-2, AD-3, AD-5	SRS-1, SRS-5, SRS-8, SRS-10, SRS-11, SRS-16, SRS-17, SRS-18, SRS-21, SRS-22, SRS-23
SG-4	Log internal task;	US-5,	URS-6,	AD-3, AD-	SRS-12, SRS-

	Post or view FAQ / issue	US-7	URS-8	5	13, SRS-17, SRS-18
SG-5	View reports / reflection	US-9, US-10	URS-9, URS-10	AD-6	SRS-19, SRS- 20

From SRS to Sequence Diagram

SRS	SQD
SRS-1	SQD-1
SRS-2	SQD-1
SRS-3	SQD-1
SRS-4	SQD-1
SRS-5	SQD-2
SRS-6	SQD-2
SRS-7	SQD-2
SRS-8	SQD-2
SRS-9	SQD-2
SRS-10	SQD-3
SRS-11	SQD-3
SRS-12	SQD-3
SRS-13	SQD-3
SRS-14	SQD-4
SRS-15	SQD-4
SRS-16	SQD-5
SRS-17	SQD-5
SRS-18	SQD-5
SRS-19	SQD-6
SRS-20	SQD-6
SRS-21	SQD-1
SRS-22	SQD-1
SRS-23	SQD-1

